

Statistical Report

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Statistical Report on Fertility Data in Singapore

Abstract

Introduction

Can time series models trained on data from 1960–2012 accurately forecast Singapore's continuing fertility decline over 2013–2024?

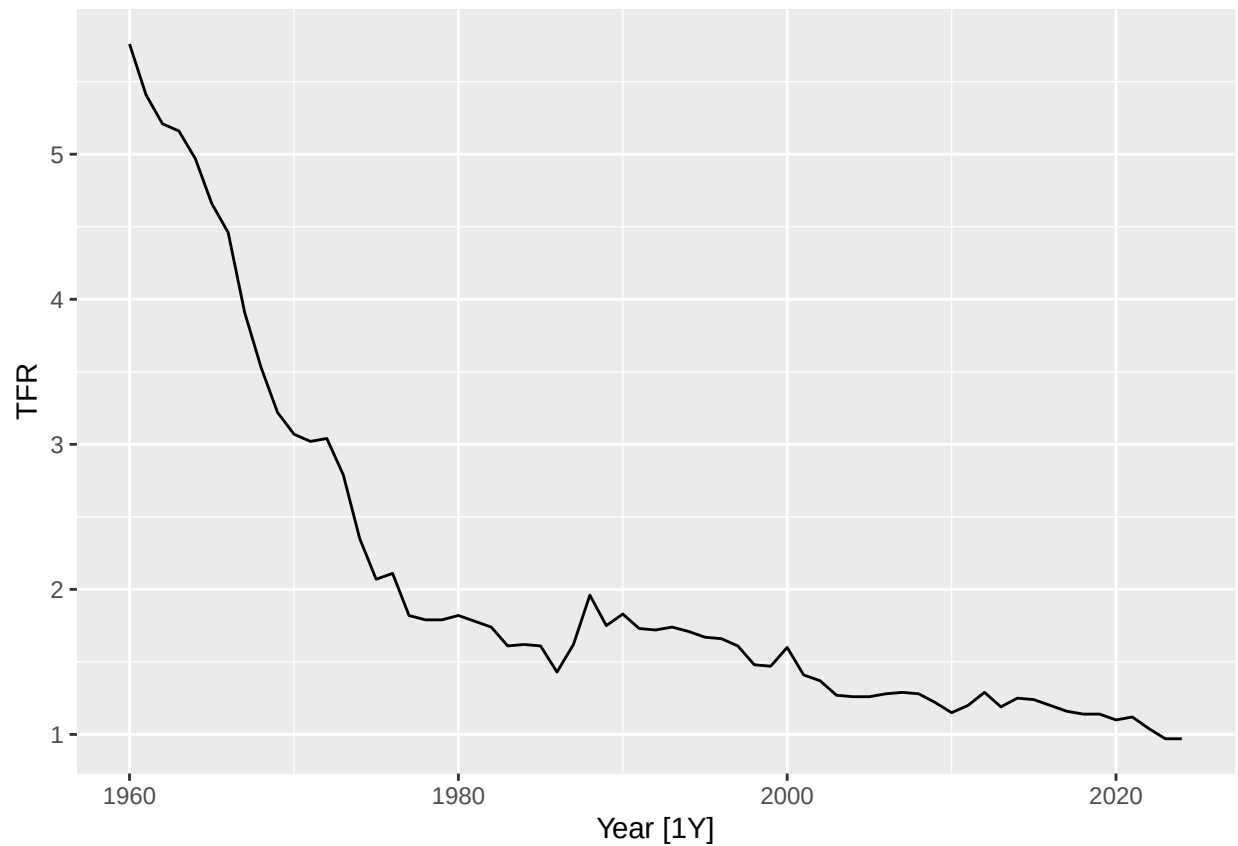


Figure 1: Total Fertility Rate in Singapore, 1960–2024

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## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

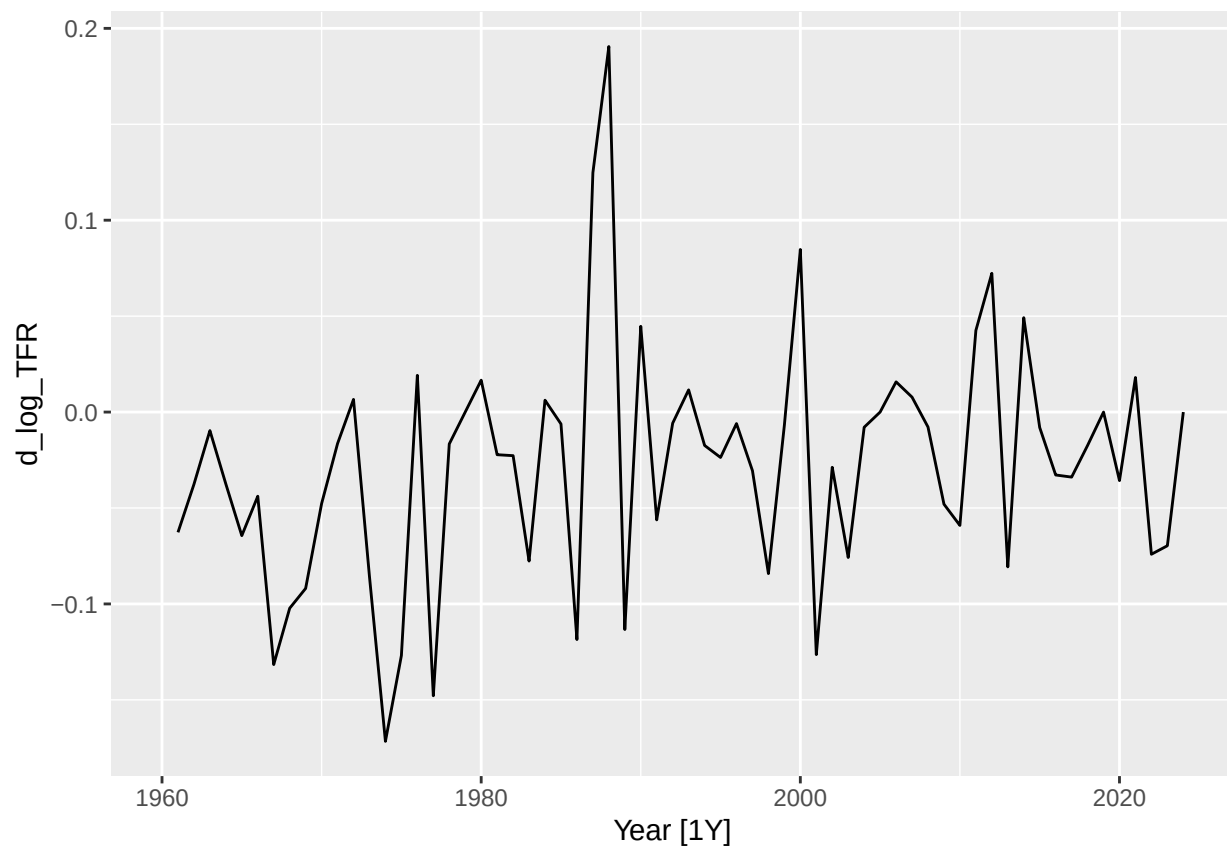
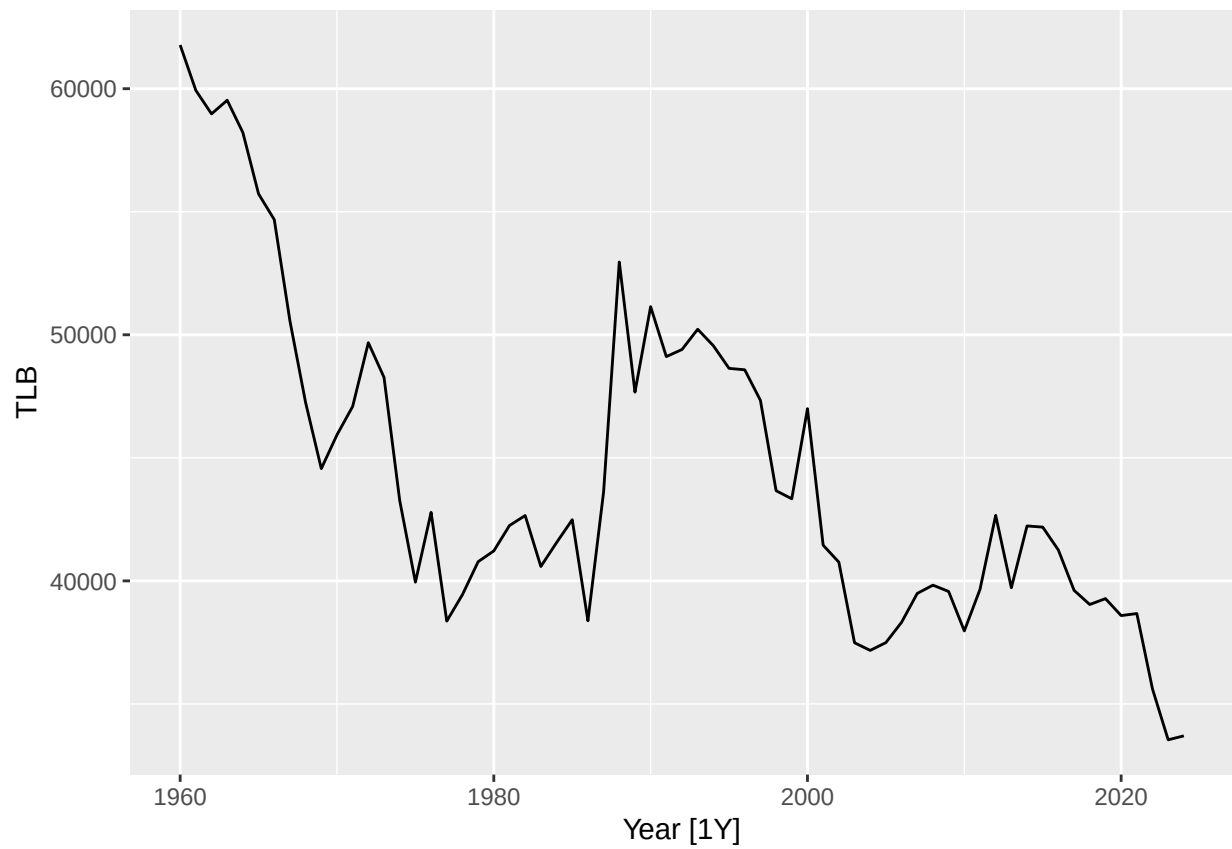
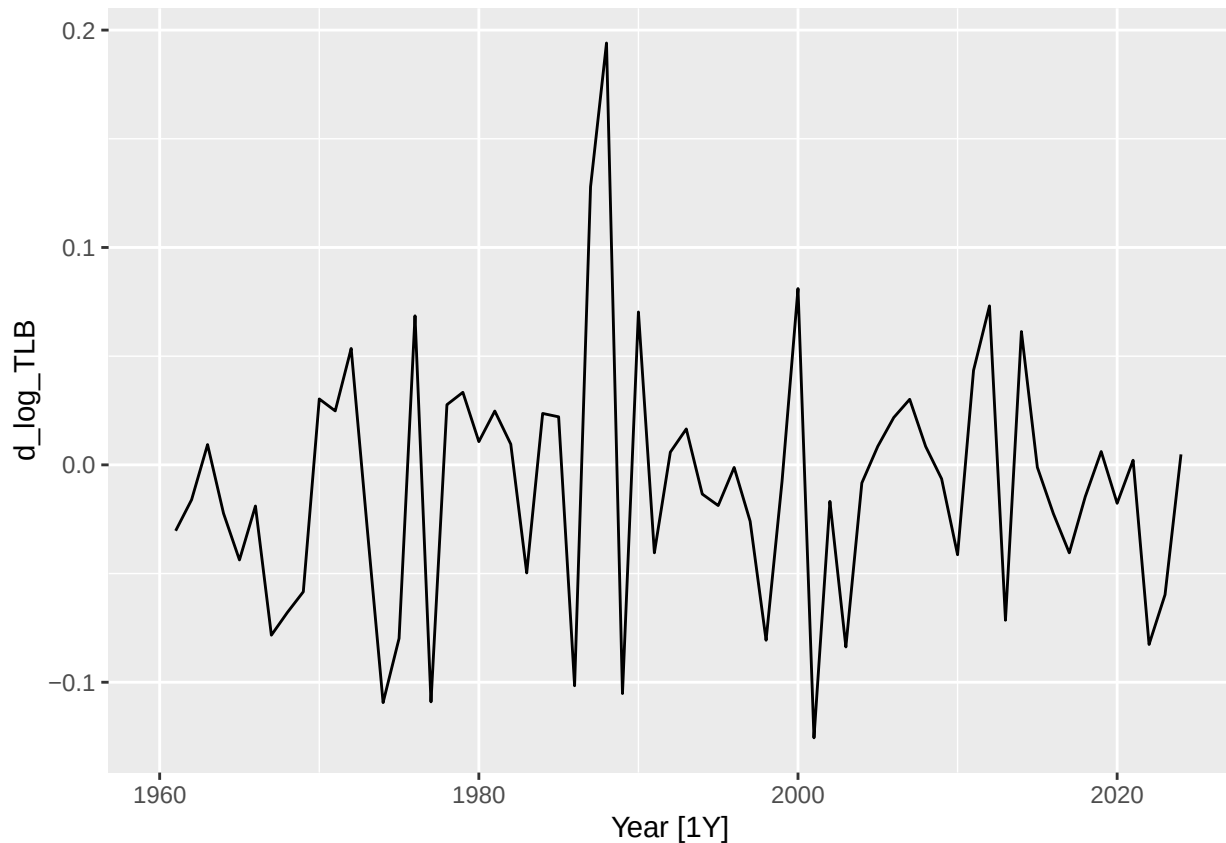


Figure 2: Total Fertility Rate in Singapore, 1960–2024



```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```



Methods

- What the data is
- talk about how the model's were chosen
 - Looked at transformation to make the data stationary
 - Then chose to log diff as it had the best __ stationarity i think
 - * The log transformation addresses **variance non-stationarity**, and the differencing addresses **mean non-stationarity** (the trend). They solve different problems. The KPSS test would confirm mean-stationarity after differencing, not “best stationarity” between log-diff and plain-diff.
 - The data still had trend for $yr=12$ (zodiac i think)
 - So informed decision to try ARMIA (12,x, y) and SARIMA models potentially

Results

Discussion

Conclusion

References

Appendices if applicable.